

THE GENERATOR IS THE TRACKER: MULTI-OBJECT TRACKING BY PAINTING PERSISTENT IDENTITY COLOURS

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ABSTRACT

Multi-object tracking (MOT) is conventionally decomposed into detection followed by association, with object identity maintained as external state — track buffers, motion models, appearance embeddings. We ask whether a video *generator* can maintain that state *in pixels*. We fine-tune a 22B text-to-video diffusion model (LTX-2.3) with a lightweight in-context LoRA to translate an RGB clip into an *ID-map* clip: a video in which every person is painted a flat, distinct colour that persists over time — same colour, same identity. Long videos are generated as chained windows, where each window is conditioned on the cleaned tail of the previous one; a brief *continuation* fine-tune teaches the model to extend a given colouring, after which identity flows through the chain with no tracker, no motion model, and no re-identification module. On the DanceTrack test server our system — to our knowledge the first generative tracker evaluated there, and the only entry with no detector and no tracking stack — reaches **40.3 HOTA**. This is well below today’s specialist state of the art (≥ 70 HOTA), but with a unique, *inverted* error profile: its association score (**AssA 44.1**) exceeds every tracker of the original benchmark suite while detection remains the sole deficit. Controlled comparisons show the mechanism matters: the same generated windows linked by classical post-hoc association score $2\times$ worse (18.2 HOTA), and frame-to-frame IoU association *fragments* tracks that the generator’s colours keep whole. On 383 mined occlusion events, the generator re-acquires identities after gaps at a 42% conditional rate where appearance-embedding baselines score zero, including gaps longer than its temporal context — evidence that the generator’s colour assignment functions as an emergent re-identification signal. We release code, checkpoints, and the full pre-registered experimental log.

1 INTRODUCTION

The dominant paradigm in multi-object tracking is *tracking-by-detection*: a detector proposes boxes per frame, and an association stage stitches them into tracks using motion models and appearance embeddings (Bewley et al., 2016; Zhang et al., 2022). Even end-to-end transformer trackers (Zeng et al., 2022) retain the same ontology — identity is *external state*, carried in query slots or track buffers alongside the video.

Video generative models suggest a different ontology. To synthesize a coherent video, a generator must implicitly solve the binding problem: it must know which pixels belong to the same object across frames, through occlusion, deformation, and camera motion. If that implicit knowledge is real, it should be extractable *as* generation. This paper tests the strongest version of that hypothesis: we ask a video diffusion model to emit, for an input RGB clip, a video of *ID-maps* — every tracked person painted a flat, distinct colour on black, with the constraint that *the same person keeps the same colour over time* (Figure 1). Identity is not a slot, an embedding, or a buffer; it is a property of the generated pixels. Decoding tracks from such a video requires only nearest-palette colour quantisation.

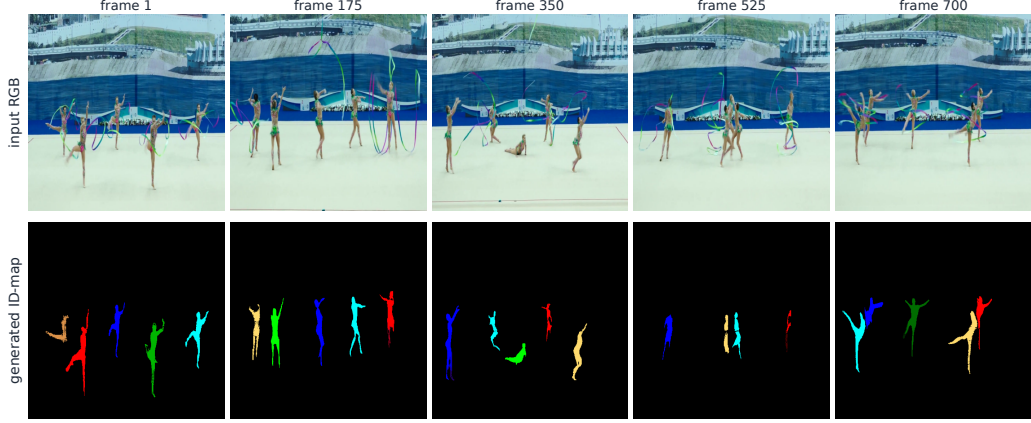


Figure 1: **Tracking by painting.** Input RGB (top) and our generated ID-map video (bottom) for a full 702-frame DanceTrack validation sequence. Identity is carried entirely by the generated colours: the same five dancers keep the same five colours across 18 chained generation windows with no tracker, no motion model, and no association step.

Realising this is not a prompt-engineering exercise. Our contributions are the mechanism and its controlled evaluation:

- **Identity-as-colour video translation.** We formalise tracking as RGB→ID-map video translation over a fixed $K=48$ CIELAB-separated palette with a lossless codec (round-trip mask IoU ≥ 0.99), and fine-tune LTX-2.3 (22B) with a rank-32 in-context LoRA — positionally aligned reference conditioning — on 40 DanceTrack training videos with SAM-derived silhouette supervision.
- **Continuation training for unbounded video length.** Windows of 49 frames are chained by conditioning each window’s first 9 frames on the *cleaned* (decoded and re-encoded) tail of the previous window. Naive inference-time chaining fails — the model fights an imposed colouring it was never trained to continue — but a brief teacher-forced *prefix* fine-tune (the target’s own first latent frames given clean with probability 0.5) teaches “continue this colouring,” raising full-video identity scores $2.7\times$.
- **The first generative tracker on the official evaluation server.** A single frozen submission achieves **HOTA 40.3 / AssA 44.1 / IDF1 45.2** — the only entry, to our knowledge, with no detector and no tracking stack. We do not claim state of the art: modern specialists reach 62–70 HOTA (Table 1). The result is the error profile, which is *inverted* relative to every specialist: association exceeds all trackers of the original benchmark suite (ByteTrack 32.1, OC-SORT 38.3, MOTR 40.2 AssA) while detection (DetA 37.6) is the sole and declared deficit.
- **The mechanism matters: controlled A/B.** On identical generated windows over the 25 full-length validation videos, generator-carried identity scores 34.8 HOTA while classical post-hoc association of the same windows scores 18.2 — the association machinery that MOT considers essential *halves* performance relative to letting the generator carry identity in pixels. A per-frame IoU tracker applied inside windows similarly fragments identities the colours keep whole (14 → 44 track fragments per window on a diagnostic video).
- **Emergent re-identification.** On 383 mined occlusion/exit events across the validation set, the generator re-binds the correct identity after the gap at a 42% conditional rate (38% for gaps of 25–48 frames), where zero-shot editing and appearance-embedding banks (VAE and DINO features) all score *zero* in our image-model baselines. Notably, gaps *longer* than the model’s 49-frame temporal context still re-acquire at 27–31%: the model re-derives the same colour from the same appearance, acting as its own re-identification function.

We report an honest failure axis as a finding: in dense scenes (>14 people) the model saturates at 10–12 painted instances per frame at 512^2 . A resolution probe shows the bottleneck is latent token

capacity — 768^2 separates crowds dramatically better — but pixel-accurate reference alignment does not transfer across resolution without resolution-matched training, which we leave as future work. Throughput is likewise not a claim: generation runs at ~ 0.8 frames per second on one A100.

2 RELATED WORK

Multi-object tracking. Tracking-by-detection pairs a detector with association by motion (Bewley et al., 2016), appearance (Wojke et al., 2017), or both at scale (Zhang et al., 2022; 2021; Zhou et al., 2020). Transformer trackers make association end-to-end but keep identity in query state (Zeng et al., 2022; Sun et al., 2020). DanceTrack (Sun et al., 2022) was designed to defeat appearance cues (uniform costumes, chaotic motion) and exposes association as the weak point of this stack: strong detectors reach DetA ~ 70 – 80 while AssA collapses to ~ 30 . We attack the association axis directly: our AssA exceeds these systems while our DetA is the honest deficit.

Perception as generation. A growing line casts perception tasks as image generation (Wang et al., 2023; Chen et al., 2022). Most directly, Gabeur et al. (2026) argue that image-generation pretraining plays the role LLM pretraining plays for language: a generalist visual learner from which individual perception tasks (segmentation, depth) are obtained by lightweight instruction tuning, outperforming specialists such as SAM (Kirillov et al., 2023). Our work is the video instantiation of this thesis, on a task an image generator cannot express: multi-object *identity* requires temporal state, and we show a video generator supplies it in pixels. To our knowledge no prior generative formulation requires *temporally persistent instance colouring* — the property that turns segmentation into tracking — nor reports on MOT leaderboards.

Video diffusion and controllable generation. We build on latent video diffusion (Rombach et al., 2022; Ho et al., 2022) and specifically LTX-Video/LTX-2 (HaCohen et al., 2024), whose in-context LoRA provides positionally aligned video-to-video conditioning (a ControlNet analogue for video), and whose native prefix conditioning enables our continuation training. Flow-matching training (Lipman et al., 2023) and LoRA adaptation (Hu et al., 2022) keep the fine-tune at $< 0.5\%$ of model parameters.

3 METHOD

3.1 IDENTITY AS COLOUR: THE ID-MAP CODEC

An *ID-map* for frame t paints instance i ’s silhouette with palette colour $c_{a(i)}$ on black, where a is a per-video assignment into a fixed palette of $K=48$ colours chosen to maximise pairwise CIELAB separation (minimum $\Delta E = 26.4$). The assignment is *sequence-stable*: one colour per identity for the whole video. Decoding inverts this by nearest-palette quantisation in CIELAB followed by per-colour connected components with a minimum-area filter; boxes are component bounding boxes and the palette index is the track id. The codec is frozen infrastructure: encode–decode round-trips preserve masks at $\text{IoU} \geq 0.99$ with 100% identity agreement, and ID-maps are stored exclusively in lossless formats (PNG / FFV1), since lossy compression smears flat colours and silently corrupts decoding.

3.2 VIDEO-TO-VIDEO FINE-TUNING WITH IN-CONTEXT LoRA

We fine-tune LTX-2.3, a 22B-parameter latent video diffusion transformer (19B video branch, 48 layers; causal video VAE with $32 \times$ spatial and $8 \times$ temporal compression), using its native *in-context* LoRA mechanism: the VAE-encoded reference video (RGB) is concatenated to the target token sequence with *aligned* positional encodings, kept clean (timestep 0), and excluded from the loss, so bidirectional self-attention learns a positionally locked translation from reference to target. We adapt only the video branch (attention QKV/out and FFN projections; rank 32; 0.4% of parameters) with a flow-matching objective. Text conditioning is a fixed prompt whose embeddings are precomputed, which drops the 24B text encoder from training memory: the entire fine-tune fits on a single 80GB A100.

Training data are 777 clips of 49 frames at 512^2 from the 40 DanceTrack training videos, with silhouette supervision produced once by box-prompted SAM (Kirillov et al., 2023) on ground-truth

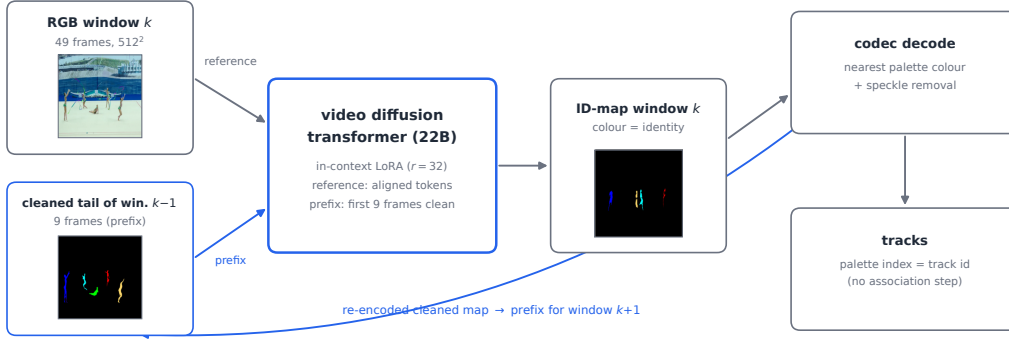


Figure 2: **System overview.** Each 49-frame window is generated by a video diffusion transformer conditioned on (i) the positionally aligned RGB reference (in-context LoRA) and (ii) a 9-frame *prefix*: the decoded, corrected, and re-encoded tail of the previous window. The generated ID-map is decoded by nearest-palette quantisation; the palette index *is* the track id, and the cleaned map seeds the next window’s prefix.

boxes (97% match rate; box fallback otherwise), and occlusion-spanning clips oversampled. The validation split (25 videos) is never used for training in any form.

3.3 CONTINUATION TRAINING: TEACHING THE MODEL TO EXTEND A COLOURING

A 49-frame window is far shorter than a video, and generating longer clips directly degrades identity (Section 4.4); full videos therefore require *chained* windows with 9-frame overlaps. The chain conditions each window’s first 9 frames — via the model’s native prefix conditioning, which anchors the first latent frames as clean tokens — on the previous window’s *cleaned* output: decoded to palette colours, speckle-filtered, and re-encoded, so the condition is always a valid ID-map rather than raw generation (a decode–correct–re-encode loop).

Applied at inference time only, this fails in an instructive way: the anchor binds the overlap frames, but past them the model reverts to its own appearance-derived colour preferences — it was trained to *invent* colourings, never to *continue* one — and errors compound down the chain. We therefore resume training with an additional intrinsic condition: with probability 0.5, the target’s own first two latent frames (9 pixel frames) are given clean and excluded from the loss, teacher-forcing the skill “propagate the colouring you were handed.” This single change lifts full-video identity scores from 0.172 to 0.466 IDF1 on our development video ($2.7\times$), while the $p=0.5$ mixture preserves unconditioned single-window quality for the chain’s first window.

3.4 INFERENCE: THE FULL-VIDEO CHAIN

Figure 2 summarises the pipeline. Given a test video, we (i) split it into 49-frame windows at stride 40; (ii) generate window k conditioned on [reference RGB_k ; prefix = cleaned tail of window $k-1$] (20 denoising steps, classifier-free guidance 4.0, spatio-temporal guidance 1.0); (iii) decode–correct–re-encode; (iv) emit boxes from owned frames (overlaps split at their midpoint), with same-colour boxes in a frame merged by union. The colour index *is* the global track id: no association step exists anywhere in the system.

4 EXPERIMENTS

Setup. We evaluate on DanceTrack (Sun et al., 2022): 40 train / 25 val / 35 test videos (700–2,400 frames each) of group dance — near-identical costumes and chaotic motion, constructed so appearance-based association fails. Metrics are HOTA/DetA/AssA (Luiten et al., 2021), IDF1, and MOTA via TrackEval; test numbers come from the official evaluation server (single submission, method frozen on validation — no test-side iteration of any kind). All fine-tuning parameters, decode

Table 1: DanceTrack **test** results (official server). Original-suite values from Sun et al. (2022) (Table 3); modern representatives — one per family — from Maggolino et al. (2023); Lv et al. (2024); Gao & Wang (2023); Zhang et al. (2023). Every baseline uses a dedicated detector; ours is the only method with no detector and no tracking stack. We do not claim state of the art: published specialists reach 69.9 HOTA (73.4 with ensembling), and entries on the live evaluation server extend above this published band. Our contribution is the paradigm and its inverted error profile: association above the entire original suite, detection the sole deficit.

	Method	Detector	HOTA↑	DetA↑	AssA↑	IDF1↑	MOTA↑
original suite	CenterTrack	✓	41.8	78.1	22.6	35.7	86.8
	FairMOT	✓	39.7	66.7	23.8	40.8	82.2
	TransTrack	✓	45.5	75.9	27.5	45.2	88.4
	ByteTrack	✓	47.7	71.0	32.1	53.9	89.6
	QDTrack	✓	54.2	80.1	36.8	50.4	87.7
	OC-SORT	✓	55.1	80.3	38.3	54.6	92.0
	MOTR	✓	54.2	73.5	40.2	51.5	79.7
modern	Deep OC-SORT	✓	61.3	82.2	45.8	61.5	92.3
	DiffMOT	✓	62.3	82.5	47.2	63.0	92.8
	MeMOTR	✓	68.5	80.5	58.4	71.2	89.9
	MOTRv2	✓	69.9	83.0	59.0	71.7	91.9
	Ours (generation only)	—	40.3	37.6	44.1	45.2	16.2

thresholds, and window geometry were pinned before each run in a pre-registered experiment log that we release. Total compute: ≈ 85 A100-hours plus one CPU-parallel evaluation pass.

4.1 DANCETRACK TEST-SERVER RESULTS

Table 1 and Figure 3 show the result. We state its scope precisely: this is a paradigm result, not a state-of-the-art result. Modern specialists reach 62–70 HOTA, and the live leaderboard extends above the published band; our 40.3 sits in the range of the 2020–2021 detector-based systems (FairMOT 39.7, CenterTrack 41.8). What no other entry offers is the profile. Ours is, to our knowledge, the only submission with no detector and no tracking stack, and the only one whose association exceeds its detection: AssA 44.1 is above every tracker of the original benchmark suite — including the end-to-end transformer MOTR (40.2) and the motion specialist OC-SORT (Cao et al., 2023) (38.3) — on the benchmark expressly built to break association, while DetA (37.6 vs. specialists’ ~ 80) is the entire deficit. Modern systems reach higher association (45.8–59.0) only through detector-boosted pipelines (Zhang et al., 2023; Gao & Wang, 2023); their margin is concentrated exactly where our diagnosis places our gap — detection — and their AssA/DetA ratio remains below one, whereas ours is above it. The generative tracker inverts the failure mode of the classical stack: it binds identity better than dedicated trackers and detects worse than dedicated detectors. Test scores exceed our validation scores (34.8 HOTA / 38.3 IDF1), consistent with a single frozen submission having no channel to overfit the test distribution.

4.2 DOES THE GENERATOR’S IDENTITY MATTER? A CONTROLLED A/B

Identity could, in principle, be recovered from generated windows by classical means. We test this with a pre-registered A/B on the 25 full-length validation videos using the *same window geometry*: (A) the chain of Section 3.4 with colour as identity; (B) windows generated *independently* (no prefix) and linked post hoc by a tuned overlap-IoU association (Hungarian matching extended to many-to-one linking, developed against an oracle-stitching diagnosis on a development video; Section 4.4).

Arm A wins by $\sim 2\times$ on every identity metric (Table 2). The post-hoc stitcher — which *tied* arm A on the 5-person development video — collapses on dense scenes, where many-to-one overlap linking over-merges crowded, identically dressed people. The generator’s colours carry through exactly the conditions that break geometric association. A stronger post-hoc method might narrow the gap, but cannot change the finding that pixels alone already outperform it.

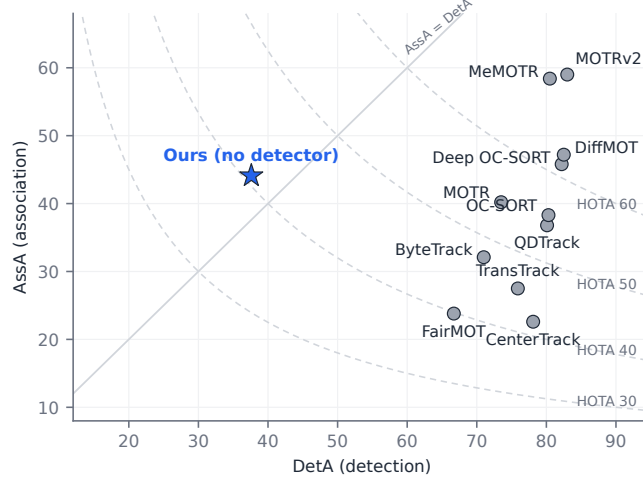


Figure 3: **The generative tracker inverts the classical error profile.** DanceTrack test decomposition: every detector-based tracker (grey) sits below the $\text{AssA} = \text{DetA}$ diagonal — detection ahead of association — while ours (star) is the only method above it, and the only one without a detector. Modern specialists (upper right) exceed our absolute scores on both axes. Dashed curves are iso-HOTA. Values from Sun et al. (2022); Maggolino et al. (2023); Lv et al. (2024); Gao & Wang (2023); Zhang et al. (2023).

Table 2: Generator-carried identity vs. post-hoc association on identical window geometry: 25 full-length validation videos, TrackEval. The classical association stack halves performance relative to reading identity off the generated colours.

Identity mechanism	HOTA $\uparrow$	DetA $\uparrow$	AssA $\uparrow$	IDF1 $\uparrow$	MOTA $\uparrow$
A: generation-carried (ours)	34.8	32.3	38.3	38.3	7.6
B: post-hoc association	18.2	19.7	17.7	17.3	−13.3

4.3 LONG-GAP RE-ACQUISITION: THE GENERATOR AS ITS OWN RE-ID

The hardest identity problem is re-acquisition: a person occluded or out of frame for dozens of frames must return with the same identity. In preliminary studies with an image-generation variant of our system, every classical remedy scored *zero* on this axis: zero-shot editing, a VAE-feature appearance bank, a DINO-feature bank, and motion-primary matching all failed to re-bind a single reappearance (appearance embeddings are uninformative on DanceTrack by design). We mined all 383 ground-truth gap events (≥ 9 absent frames, then reappearance) across the 25 validation videos and measured, for events where the person was tracked before the gap and re-detected after it, whether the identity matched (pre/post majority id over 15 visible frames at IoU 0.3).

Two mechanisms are visible in Table 3 (a qualitative example in Figure 4). For gaps *inside* the temporal context (9–48 frames), attention sees through the occlusion, and the chained model re-binds at 38–47% — the capability that motivated using a video model. For gaps *longer* than the context (49–150 frames), where no attention path exists, re-acquisition persists at 27–31%: the model tends to re-derive the *same* colour for the *same* appearance. The generator’s colour assignment is an emergent re-identification function — precisely the claim that generators learn identity-bearing features.

4.4 ABLATIONS AND ANALYSIS

Continuation training is necessary. With the same chain at inference but no prefix training, full-video IDF1 on the development video is 0.172 — *below* independent windows — because the model

Table 3: Conditional re-acquisition rate after ground-truth gaps (383 events, validation set). The 49-frame window is the model’s temporal context. Image-model baselines (appearance banks, zero-shot editing) score 0 across all bins.

Gap length (frames)	9–24	25–48	49–150	all
n (conditional events)	127	50	26	205
Ours (chained)	0.472	0.380	0.269	0.420
Independent windows + association	0.385	0.328	0.310	0.361
Appearance-bank baselines	0	0	0	0

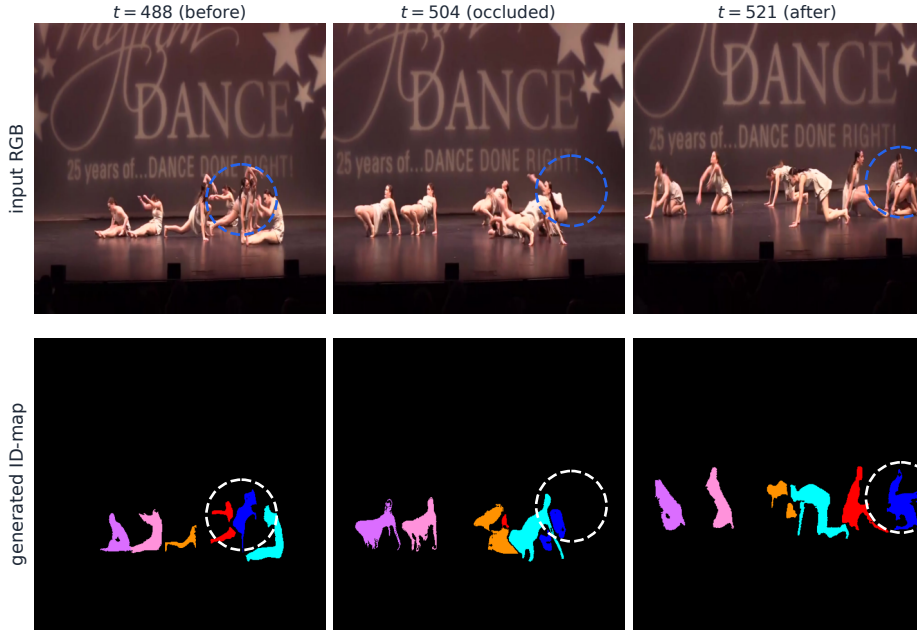


Figure 4: **Re-acquisition through occlusion.** A dancer (dashed circle) is fully occluded for 27 ground-truth frames and returns painted the *same blue* — the generator, not an association module, re-binds the identity. During the occlusion the model even maintains a partial hypothesis fragment.

overrides the imposed colouring mid-window and errors compound. Prefix training lifts it to 0.466 while single-window validation quality is preserved (foreground IoU 0.589 vs. 0.605).

Longer windows are not the answer. Generating 121-frame clips with the 49-frame-trained model keeps mask quality but degrades identity sharply (IDF1 0.549 over frames 1–49 vs. 0.312 over frames 50–121; colour-switch rate $2.3\times$): identity persistence does not extrapolate beyond the trained horizon, motivating chaining at the trained length.

The generator’s colours beat frame-level geometric association. An oracle decomposition on the development video shows window-level identity is strong (per-window IDF1 0.661; a perfect post-hoc sticher would reach 0.698). Replacing colour identity with per-frame IoU linking *inside* windows fragments 14 colour-consistent units per window into 44 tracklets — fast dance motion breaks frame-to-frame IoU where generated colour remains stable. Colour is the more reliable identity signal precisely where classical tracking is weakest.

Dense scenes are token-capacity-bound (Figure 5). Failure concentrates in crowded videos (14–25 people: IDF1 0.18–0.33; 4–9 people: 0.40–0.86). The model saturates at 10–12 painted instances per frame at 512^2 (where a frame is 16×16 latent tokens) regardless of ground-truth count, while sparse-scene recall is 0.86–1.0 — people are not small, they are unrepresentable in that token budget. Regenerating a dense video at 768^2 separates crowds dramatically (merge rate halves; visible

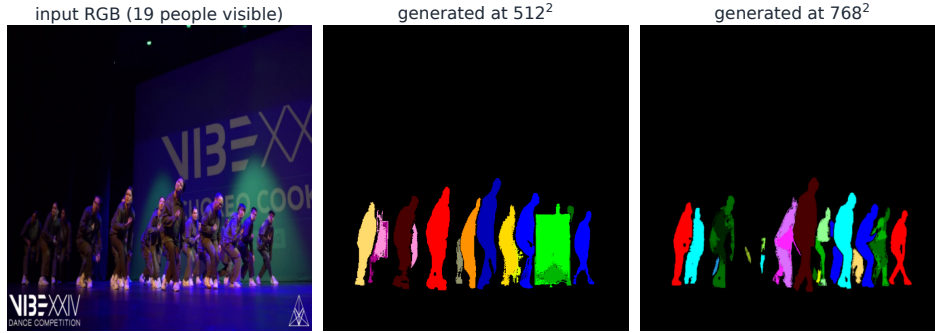


Figure 5: **Dense scenes are token-capacity-bound.** At 512^2 (16×16 latent tokens per frame) the model paints only ~ 12 of 19 people, merges neighbours, and hallucinates a non-person region; at 768^2 , with the *same* checkpoint, crowd separation improves dramatically — but reference alignment warps at the unseen resolution, so boxes degrade. Resolution-matched training is future work.

instances rise) but box-level accuracy *drops*: the 512-trained LoRA’s reference alignment warps nonuniformly at unseen resolution. Resolution-matched training is the indicated fix and is left as future work.

5 DISCUSSION

Our results provide task-level evidence for the position of Gabeur et al. (2026) that strong generative models are generalist vision learners: one pretrained generator, plus a task-specific fine-tune measured in tens of GPU-hours and $<0.5\%$ of parameters, replaces an entire task-specific stack (detector, motion model, re-identification, association logic). Tracking is a particularly sharp test of the thesis because it is not a per-frame labelling task — it requires persistent state — and the video generator’s pretraining evidently supplies exactly that: temporal attention re-binds identities through occlusions (Section 4.3), and the learned appearance $\rightarrow$ colour mapping acts as an emergent re-identification function beyond the attention horizon. We therefore expect this recipe to inherit the scaling of its foundation: our two dominant deficits — crowd detection bound by latent token capacity, and throughput — are properties of the 2026-era backbone and its resolution, not of the formulation, and should improve with each generation of video-generative foundation models.

6 LIMITATIONS

Detection in dense scenes (DetA 37.6) and throughput (~ 0.8 fps on one A100; a 22B model denoises every window in 20 steps) are the two honest deficits, and we make no speed claims. The palette bounds simultaneous identities at $K-1 = 47$. The evaluation is single-domain: DanceTrack isolates the association axis by construction, and transfer to appearance-diverse domains (e.g. MOT17) at full scale remains untested — our preliminary image-model probes there showed the same detection-bound behaviour. Continuation training teacher-forces ground-truth prefixes, leaving a train/test gap to self-generated (cleaned) prefixes; mild quality erosion deep in long chains is visible.

7 CONCLUSION

Identity can live in pixels. A video generator, lightly fine-tuned to paint people with persistent colours and to *continue* a colouring it is handed, tracks full-length videos on the official DanceTrack server with no detector, no motion model, no re-id module, and no association step — and with association quality above the entire original benchmark suite, on the benchmark built to break association. Modern detector-based specialists score higher in absolute terms; none share this error profile, and none dispense with the tracking stack. Controlled comparisons show that this is not a curiosity of output format: the same generated evidence, linked classically, is twice as bad, and the generator re-acquires identities through occlusions that defeat appearance embeddings entirely. We

offer this as evidence that video generators learn identity-bearing object representations, and that “generation as the interface” is a viable, measurable path for video perception. The open deficits — crowd detection bound by latent token capacity, and throughput — are engineering axes with visible levers, not conceptual ones.

REPRODUCIBILITY STATEMENT

All experiments are logged in a pre-registered experiment record (config hashes, seeds, costs, and gate verdicts per run) released with the code, ID-map codec, training and inference configurations, and evaluation scripts. The fine-tune uses a single 80GB GPU; total compute for all reported results is ≈ 85 A100-hours. The DanceTrack test result is a single frozen submission to the official server, with the submission archive and per-sequence outputs included in the release.

ETHICS STATEMENT

This work uses only the public DanceTrack benchmark under its terms. Person tracking has surveillance applications; our system offers no capability beyond existing trackers’ accuracy envelope and is trained on a single choreographed-dance domain. We discuss failure modes (dense crowds) explicitly to discourage overestimation of capability.

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A LLM USAGE DISCLOSURE

Large language models were used substantially in this research: an LLM coding assistant (Anthropic Claude) implemented experiment scripts, orchestrated training and evaluation runs. Research direction, hypotheses, gate decisions, spending approvals, and final claims were set and verified by the human authors. All reported numbers originate from logged experimental runs, not from LLM generation.

B PINNED HYPERPARAMETERS AND ADDITIONAL DETAILS

Model. LTX-2.3 dev checkpoint (22B; 19B video transformer, 48 layers); Gemma-3 text encoder used only to precompute one fixed prompt embedding. **LoRA.** Rank 32, α 32, video-branch attention (QKV, output) and FFN projections only; audio branch excluded. **Training.** Phase A (translation): 1,500 steps, $\text{lr } 2 \times 10^{-4}$, bf16, gradient checkpointing, flow matching with shifted-logit-normal

timestep sampling, batch 1, 777 preprocessed clips ($512^2 \times 49$). Phase B (continuation): resume to step 2,500 with intrinsic prefix condition (2 latent frames, $p = 0.5$); best checkpoint selected by in-loop validation on all 25 validation videos (foreground IoU). **Inference.** 49-frame windows, stride 40 (9-frame overlap = prefix length); 20 steps; CFG 4.0; STG 1.0; seed 0; 512^2 . **Decode.** $K=48$ palette, index 0 = background; nearest-palette in CIELAB; per-colour connected components; min area 150 px; per-frame same-colour union. **Windows to tracks.** Overlap ownership split at midpoint; colour index = track id.